PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

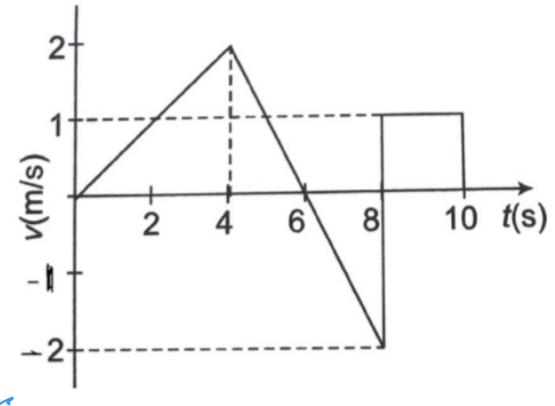
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18. The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a, b and c are constants . The dimensions of a, b and c are, respectively ,

A) $[LT^{-2}], [L] \text{and} [T]$ **B)** $[L^2], [T] \text{and} [LT^2]$

C) $[LT^{-2}], [LT] \text{and} [L]$ **D)** $[L], [LT] \text{and} [T^2]$

19. The velocity –time graph of a body moving in a straight line is shown. The displacement of the body in 10 s is .



$S = \left(\frac{1}{2} \times 6 \times 2\right) - \left(\frac{1}{2} \times 2 \times 2\right) + (2 \times 1)$

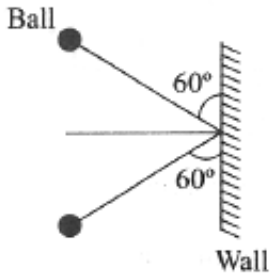
$= 6 - 2 + 2 = 6$

A) 4 m **B)** 6 m **C)** 8 m **D)** 10 m

20. A ball of mass 3 kg moving with a speed of 100 m/s , strikers a wall at an angle 60^0 as shown in the figure. The ball rebounds at the same speed and remains in contact with the ball for 0.2 s. The force exerted by the ball on the wall is

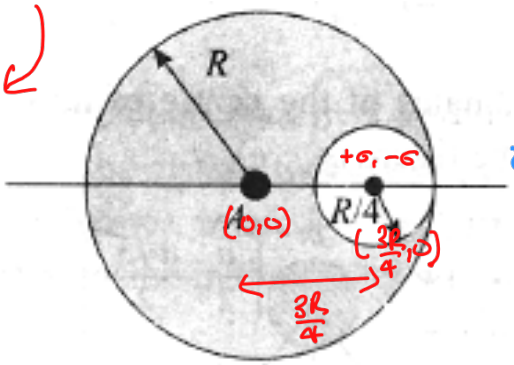
$F_{avg} = \frac{2mv\sin\theta}{\Delta t} = \frac{2 \times 3 \times 100 \times \sin 60^\circ}{0.2}$

$= 3000 \times \frac{\sqrt{3}}{2} = 1500\sqrt{3}\ N$



A) $1500\sqrt{3}N$ **B)** $1500\ N$ **C)** $300\sqrt{3}N$ **D)** $300N$

21. The centre of mass of the shaded portion of the disc is (The mass is uniformly distributed in the shaded portion)



A) $\frac{R}{20}$ to the left of A **B)** $\frac{R}{12}$ to the left of A

C) $\frac{R}{20}$ to the right of A **D)** $\frac{R}{12}$ to the right of A

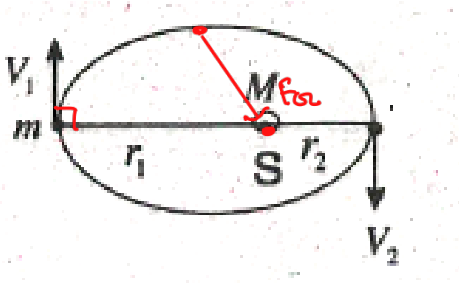
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

22. A planet of mass M moves along an ellipse around then sun, so that its maximum and minimum distances from the sun are r_1 and r_2 respectively. Then



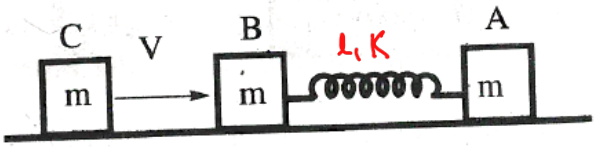
A) $V_1 r_1 = V_2 r_2$

B) Total energy of plant is always a constant

C) Angular momentum of plant is $m\sqrt{\frac{2GM(r_1 r_2)}{(r_1 + r_2)}}$

D) $V_2 r_1 = V_1 r_2$

23. Two blocks A and B each of mass m are connected by a massless spring natural length ' l' and spring constant K. The blocks are initially resting on a smooth horizontal floor with the spring at its natural length as shown. A third identical block C also of mass m, moves on the floor with a speed v along line joining A and B and collides elastically with B, then



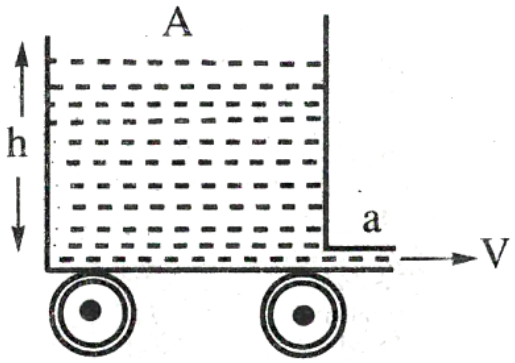
A) The K.E of the A-B system, at the maximum compression of the spring is zero .

B) The K..E of the system, at the maximum compression of the spring is $mv^2/4$.

C) The maximum compression of the spring is $v\sqrt{\frac{m}{K}}$

D) The maximum compression of the spring is $v\sqrt{\frac{m}{2K}}$

24. A container having a hole at the bottom is free to move on a frictionless horizontal surface. As the liquid comes out. Container moves in backward direction with an acceleration α and finally acquired velocity v (When all liquid drains out) Neglect the mass of container. The correct options are



A) α is independent of height of liquid in vessel

B) α depends only A and a and g

C) v depends upon α and h

D) v independent of α

(22) $mv_1r_1 = mv_2r_2$

$v_1r_1 = v_2r_2$ — (1) $\Rightarrow v_2 = \frac{v_1r_1}{r_2}$

From COME

$\frac{1}{2}mv_1^2 - \frac{GMm}{r_1} = \frac{1}{2}mv_2^2 - \frac{GMm}{r_2}$

$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = GMm\left(\frac{1}{r_2} - \frac{1}{r_1}\right)$

$\frac{1}{2}m\left(v_2^2 - v_1^2\right) = GMm\left(\frac{r_1 - r_2}{r_1r_2}\right)$

$\left(\frac{v_1r_1}{r_2}\right)^2 - v_1^2 = 2GM\left(\frac{r_1 - r_2}{r_1r_2}\right)$

$v_1^2\ \frac{r_1^2 - r_2^2}{r_2^2} = 2GM\ \frac{r_1 - r_2}{r_1r_2}$

$v_1^2\ \frac{(r_1 - r_2)(r_1 + r_2)}{r_2^2} = 2GM\ \frac{(r_1 - r_2)}{r_1r_2}$

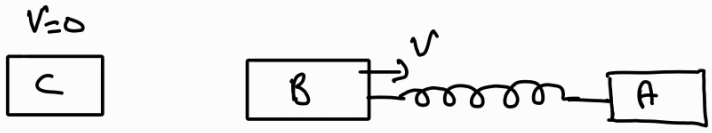
$v_1^2 = \frac{2GMr_2}{r_1(r_1 + r_2)}$

$v_1 = \sqrt{\frac{2GMr_2}{r_1(r_1 + r_2)}}$

$L = mv_1r_1 = m\sqrt{\frac{2GMr_2}{r_1(r_1 + r_2)}}\ r_1$

$= m\sqrt{\frac{2GMr_1r_2}{(r_1 + r_2)}}$

(23)



at maximum compression

$v_A = v_B$

From COLM

$mv = mv_A + mv_B$

$v_A = v_B = v/2$

$\therefore K = \frac{1}{2}m\left(\frac{v}{2}\right)^2 \times 2 = \frac{mv^2}{4}$

From COME

$$\frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{v}{2}\right)^2 \times 2 + \frac{1}{2}kx_{max}^2$$

$$mv^2 - \frac{mv^2}{2} = kx_{max}^2$$

$$\frac{mv^2}{2} = kx_{max}^2$$

$$x_{max} = v\sqrt{\frac{m}{2k}}$$

(24) $\frac{dv}{dt} = av$

$\frac{dm}{dt} = \rho av$

$F_{avg} = v\frac{dm}{dt} = \rho av^2$

$\alpha = \frac{\rho av^2}{m} = \frac{\rho av^2}{Ah\rho}$ $\quad (v = \sqrt{2gh})$

$\alpha = \frac{a \times 2gh}{Ah}$

$\alpha = \frac{2ag}{A}$

time taken to empty the tank, $t_0 = \frac{A}{a}\sqrt{\frac{2h}{g}}$

$v = u + at$

$= 0 + \alpha t_0$

$v = \frac{2ag}{A} \times \frac{A}{a}\sqrt{\frac{2h}{g}}$

$= 2\sqrt{2gh}$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

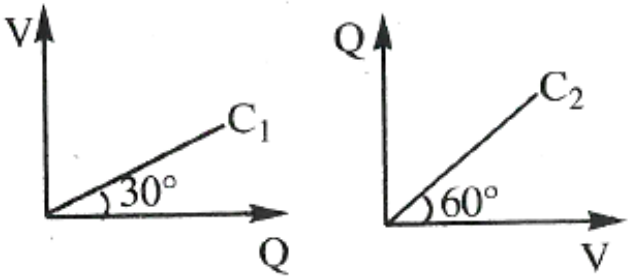
Full Marks: +4If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

25. Tuning fork A of frequency 258 Hz gives 8 beats with a tuning fork B. When the tuning fork A is filed and again A and B sounded the number of beats decreases. The frequency of B is ……

26. A heavy wheel of radius 20 cm and weight 10 kg is to be dragged over a step of height 10 cm, by a horizontal force F applied at the centre of the wheel . The minimum value of " F " is (in kgwt) ……..

27. The refractive indices of crown glass prism for C,D and F lines are 1.527, 1.530 and 1.535 respectively. Then the dispersive power of the crown glass prism is ……. (round off nearest integer)

28. Two different capacitors with capacity C_1 and C_2 are being charged. Graphs for C_1 and C_2 are as follow. Find the value of C_1 / C_2.



29. The speeds of 5 molecules of gas (in arbitrary units) are as follows 2,3,4,5,6. The root mean square speed for these molecules is ……..

30. when temperature of a gas is 20^0C and pressure is changed from $P_1 = 1.01\times10^5\,Pa$ to $P_2 = 1.165\times10^5\,Pa$ then volume changed by 10 %. The bulk modulus is $n\times10^5$ Pa then n is...

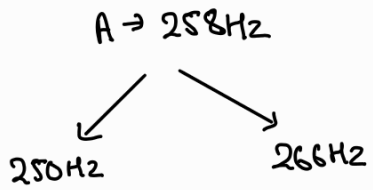
SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated <u>according to the following marking scheme:</u>
- ***Full Marks:* +3** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

Paragraph-I

The thermal coefficient of resistivity is given $\alpha = \frac{1}{\rho}\frac{dp}{dT}$ where ρ is resistivity at a temperature T. α is not constant, rather varies as $\alpha = \frac{-n}{T}$, where T is temperature in Kelvin and n is a constant . For carbon $\alpha = -0.0005K^{-1}$ and $\rho = 3.5\times10^{-5}(\Omega - m)$ at 27^0C

(25)

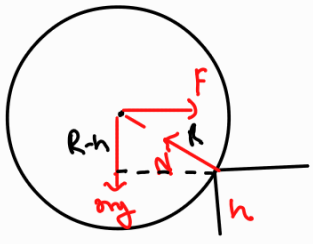


on filing $f_A \rightarrow \uparrow$

& $f_{Beat} \rightarrow \downarrow$

$\Rightarrow f_B = 266\,Hz$

(26)



Torque about Step

$\tau_F \geq \tau_{mg}$

$\Rightarrow F(R-h) \geq mg\sqrt{R^2-(R-h)^2}$

$\Rightarrow f \geq \frac{mg}{(R-h)}\sqrt{R^2-(R-h)^2}$

$f_{min} = \frac{10\times10}{(20-10)} \times \sqrt{(20)^2-(20-10)^2}$

$= \frac{100}{10} \times \sqrt{400-100}$

$= 10\times\sqrt{300}$

$= 100\times\sqrt{3}$

$\approx 173\ N$

$\simeq 17.3\ Kg\ wt$

$\simeq 17\ Kg\ wt$

(27) $\omega = \frac{\mu_F - \mu_C}{\mu_D - 1} = \frac{1.535-1.527}{1.530-1} = \frac{0.008}{0.530} \approx 0$

(28) $Q = CV \Rightarrow V = \frac{1}{C}Q$

$\frac{1}{C_1} = \tan 30^\circ = \frac{1}{\sqrt{3}} \Rightarrow C_1 = \sqrt{3}$

$C_2 = \tan 60^\circ = \sqrt{3}$

$\frac{C_1}{C_2} = \frac{\sqrt{3}}{\sqrt{3}} = 1$

(29) $v_{rms} = \sqrt{\frac{2^2+3^2+4^2+5^2+6^2}{5}} = \sqrt{\frac{4+9+16+25+36}{5}}$

$= \sqrt{\frac{90}{5}} = \sqrt{18} \simeq 4$

(30) $\frac{\Delta V}{V} \times 100 = 10 \Rightarrow \frac{\Delta V}{V} = 0.1$

$\beta = \frac{\Delta P}{\frac{\Delta V}{V}} = \frac{1.165 \times 10^5 - 1.01 \times 10^5}{0.1} = 0.155 \times 10^5 = 1.55 \times 10^5$

$\simeq 2 \times 10^5$

(31) $\alpha = \frac{1}{S}\frac{dS}{dT}$

$-0.0005 = \frac{1}{3.5 \times 10^{-5}} \frac{dS}{dT}$

$\frac{dS}{dT} = -5 \times 10^{-4} \times 3.5 \times 10^{-5}$

$= -17.5 \times 10^{-9}$

$= -1.75 \times 10^{-8}$

$\Rightarrow -\frac{n}{T} = \frac{1}{S}\frac{dS}{dT}$

$\Rightarrow -n \int \frac{dT}{T} = \int \frac{dS}{S}$

(32) $\alpha = -\frac{n}{T}$

$\pm 0.0005 = \pm \frac{n}{300}$

$n = 0.0005 \times 300$

$= 0.1500$

≈ 0

(33) Current through $R_1 = 0$, because circuit is broken.

(34) In Steady State

$$R_{eq} = R_1 + \frac{R_2 R_3}{R_2 + R_3} = 10 + \frac{20 \times 30}{20+30} = \frac{500+600}{50} = \frac{1100}{50} = 22\Omega$$

$$I = \frac{50}{22}$$

Current through R_2

$$i_2 : i_3 = \frac{1}{R_2} : \frac{1}{R_3} = \frac{1}{20} : \frac{1}{30} = 3:2$$

$$i_2 = \frac{3}{5} \times \frac{50}{22} = \frac{30}{22} = \frac{15}{11}$$

$$i_3 = \frac{2}{5} \times \frac{50}{22} = \frac{10}{11}$$

31. The value of $\frac{d\rho}{dT}$ is $-x \times 10^{-8} \Omega m / K$, find x (round off nearest integer)

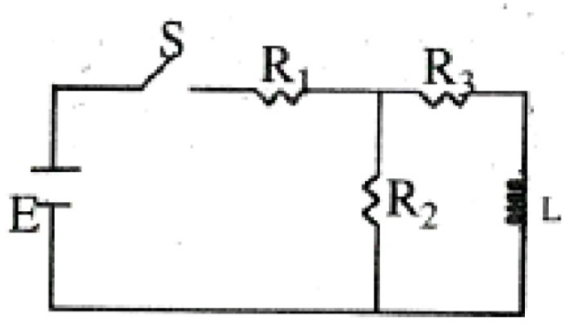
32. Find the value of ' n' for carbon (round off nearest integer)

Paragraph-II

$E = 50V$, $R_1 = 10\Omega$

$R_2 = 20\Omega$, $R_3 = 30\Omega$

and L=2mH



33. After the steady state is attained , current through R_1 immediately after S is reopened is

34. After the steady state is attained, current through R_2 immediately after S is reopened is nearly (round off nearest integer)

PHYSICS

18	**A**	19	**B**	20	**A**	21	**A**	22	**ABC**	23	**BD**
24	**ABC**	25	**266**	26	**17**	27	**0**	28	**1**	29	**4**
30	**2**	31	**2**	32	**0**	33	**0**	34	**1**		